

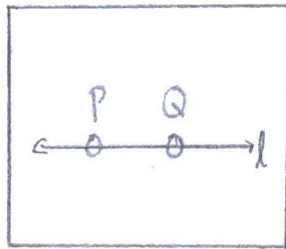
Chapter 7: Independence of the Parallel Postulate

K-exercises:

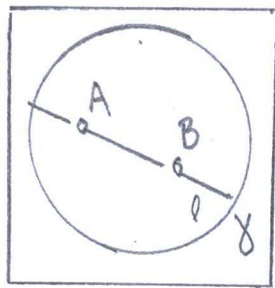
K-1.

Klein Model verified:

Incidence Axiom #1: For every point P and

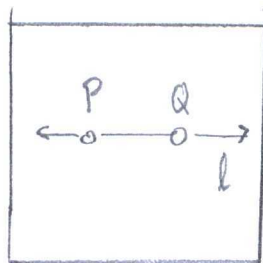


for every point Q not equal to P , there exists a unique line $\overleftrightarrow{l}$ incident with P and Q .

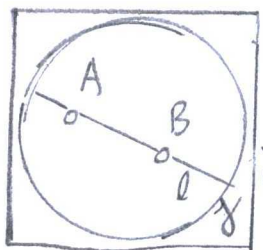


Two distinct points A and B interior to circle X have a unique open chord l in X with both A and B .

Incidence Axiom #2: For every line $\overleftrightarrow{l}$, there

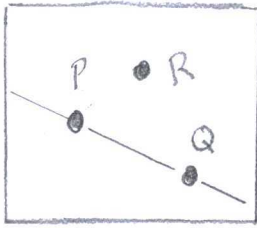


exists at least two distinct points incident with $\overleftrightarrow{l}$.

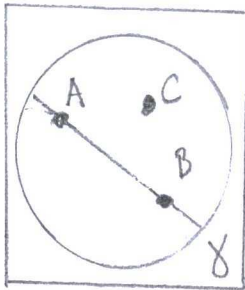


A chord in circle X is incident to points A and B .

Incidence Axiom #3: There exists three

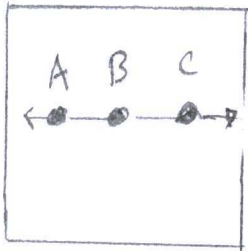


distinct points with the property that no line is incident with all three of them.

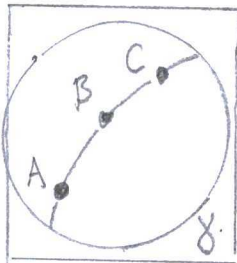


Three unique points exist: A, B, C in circle X , but a chord cannot be incident with all three.

Betweenness Axiom #1: If $A * B * C$, then A, B , and

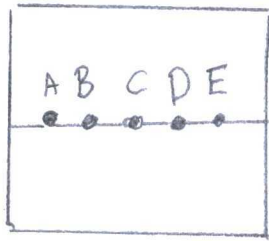


C are three distinct points all lying on the same line, and $C * B * A$.

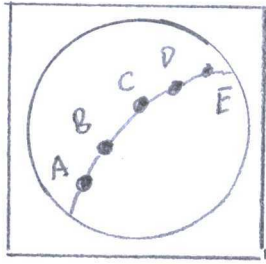


A chord $A)(B=A)(C$ when $A * B * C$ in a circle X . The chord $C)(B=C)(A$ is the same, and $C * B * A$.

Betweenness Axiom #2: Given any two distinct

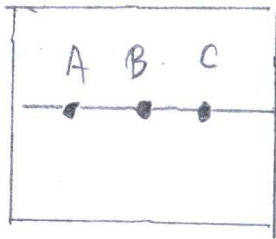


points B and D , there exists A , C , and E lying on $\overleftrightarrow{BD}$ such that $A*B*D$, $B*C*D$, and $B*D*E$

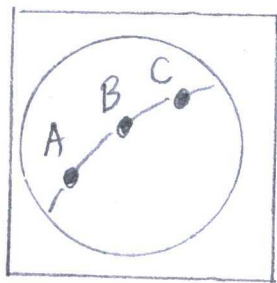


By any two distinct points B and D , three other points are on chord $B(D)$ such $A*B*D$, $B*C*D$, and $B*D*E$.

Betweenness Axiom #3: If A , B , and C are

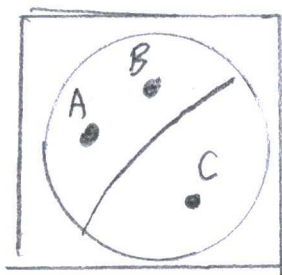


three distinct points on the same line, then only one of the points is between the two.



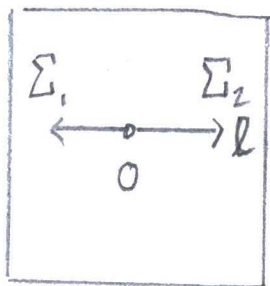
On chords, if three distinct points, then only one of the points is between the two.

Betweenness Axiom #4: For every line and for any three points A , B , and C



(ii) If A and B are on opposite sides of a chord, and if B and C are on opposite sides of a chord, then A and C are on the same side of the chord.

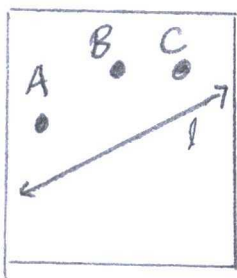
Dedekind's Axiom (pg 134): Suppose the set



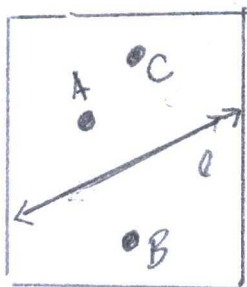
$\{l\}$ of all points on a line l is the disjoint union $\Sigma_1 \cup \Sigma_2$ of two non-empty subsets such no point of either subset is between two points of the other. Then there exists a unique point O on l such one of the subsets is equal to a ray of l with vertex O and the other subset is equal to the complement.

For a chord in X , the

not lying on $\vec{l}$:

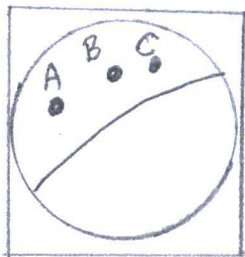


(i) if A and B are on the same side of $\vec{l}$ and if B and C are on the same side of $\vec{l}$, then A and C are on the same side of $\vec{l}$.



(ii) If A and B are on opposite sides of $\vec{l}$ and if B and C are on opposite sides of $\vec{l}$, then A and C are on the same side of $\vec{l}$.

For every chord and three points A, B, and C not on the chord:



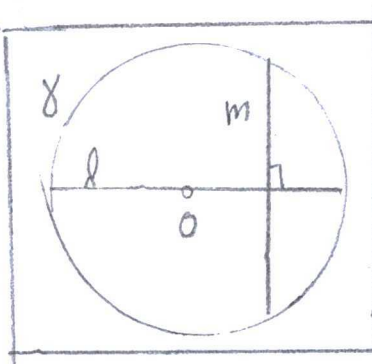
(i) if A and B are on the same side of a chord, B and C the same side of the chord, then A and C are on the same side of the chord.

Set $\{\vec{l}\}$ of all points on the chord is a disjoint union $\Sigma_1 \cup \Sigma_2$ of two non-empty subsets such no subset is between the other. Then there exists a unique point O on $\vec{l}$ such one of the subsets is equal to the ray of $\vec{l}$ with vertex O and the other subset is equal to the complement.

K-2

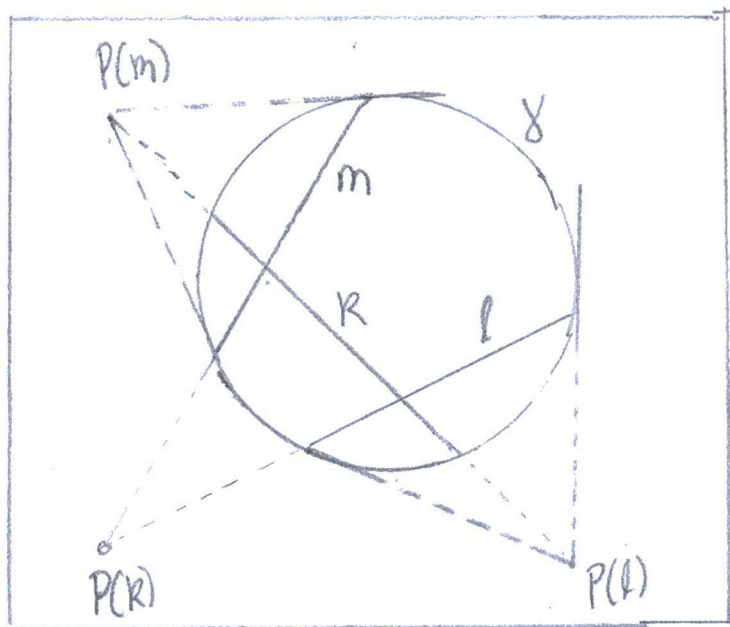
a) (pg 308) "Perpendicularity in the Beltrami-Klein Model"

A diagram showing the common perpendicular k to l and m in the Klein model.



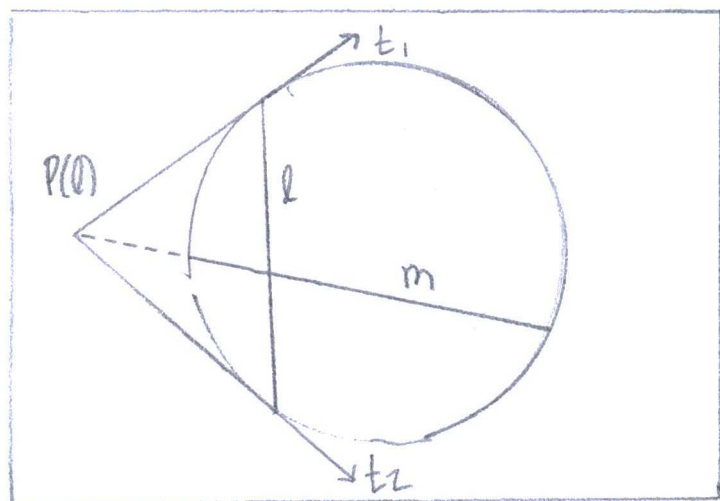
"one of l and m is a diameter. $\therefore l \perp m$."

b) (from exercise) "Two lines in Klein



model that are perpendicular to l and asymptotically parallel to m ."

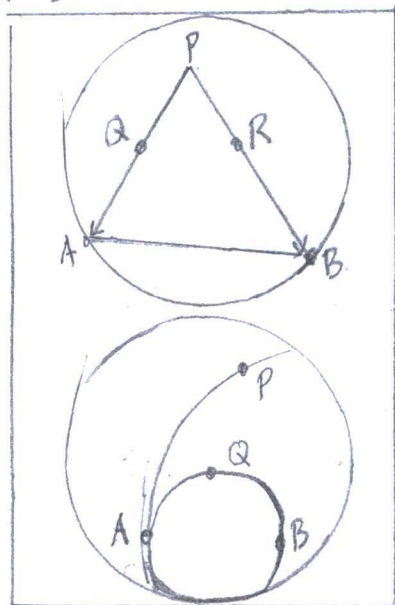
c) (from exercise) "Three parallel lines



in the Klein model that do not have a common transversal."

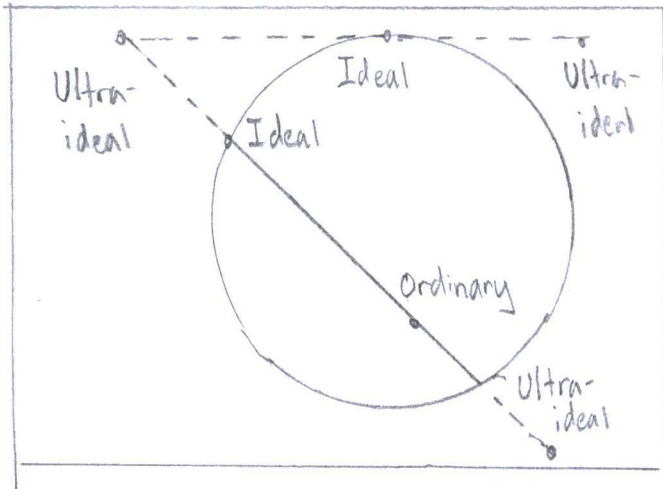
K-3

a) "Parallels limit rays in Klein



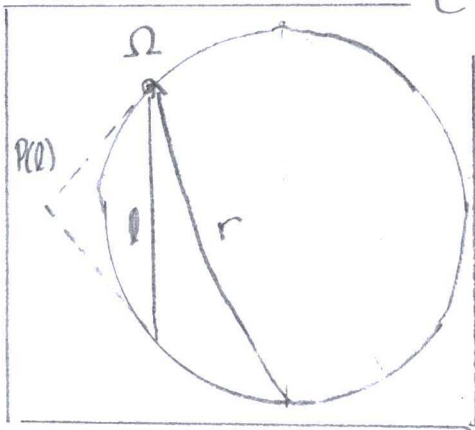
models. If ordinary point Q and ideal point A form a line. Then another ideal point B forms a Klein line AB . $\overline{PA}$ limits line AB ."

b) (from exercise) "The only case



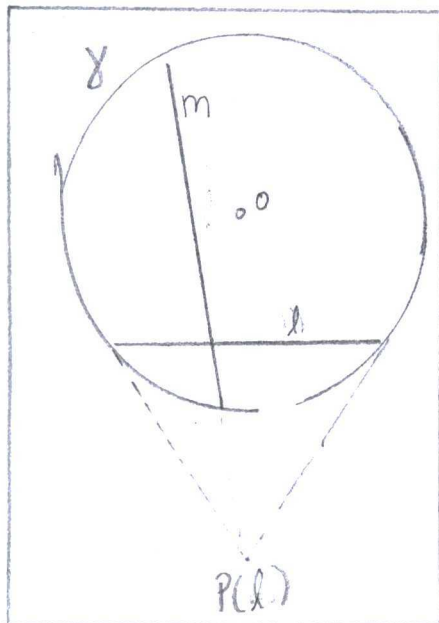
Which there is a Klein line joining $P(l)$ and $P(m)$ is, when lines $\vec{l}$ and $\vec{m}$ are divergently parallel."

c) An ultra-ideal point $P(l)$ forms a line l at ideal point Ω and γ . Line l limits parallels by ray r at Ω .



K-4

(from K-exercise) "The line extending m passes through the pole of l "



Proof Hypothesis

Evidence

1) $\overline{OB} = \overline{AB}$ is the diameter of γ

Euclid Postulate III

2) Circle γ cuts

Proposition 2.1

$\overline{AB}$ in points

C and D.

3) Points T and

Proposition 2.1

"Klein disc

- without orthogonal circle."

U are the points of intersection to γ and δ .

4) Circles γ and δ are tangent because $(\overline{OT})^2 = (\overline{OC})(\overline{OD})$

or $(\overline{OU})^2 = (\overline{OC})(\overline{OD})$

5) Circles γ and δ are orthogonal.

6) Line m is $\overline{CD}$ and l is $\overline{TU}$

7) m passes through the pole of l at O

8) An inverse circle δ' exists from point B

9) l passes through the pole of m at O'

Lemma 7.1

Proposition 7.5

Axiom C-2

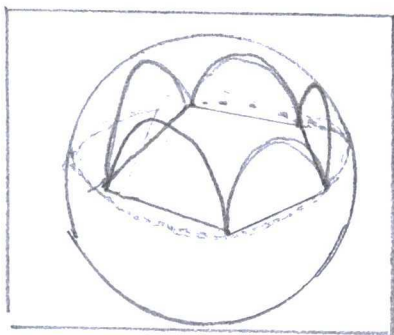
(pg 314) "Definition-Pole"

Proposition 7.3

Corollary to Proposition 7.4.

K-5

(from K-exercise) "In the hyperbolic plane there exists a pentagon with five right angles."



"Klein model...
Pentagon"

Equation to Internal Angle by
law of cosines:

$$|\overline{TP}|^2 + |\overline{TQ}|^2 - 2|\overline{TP}||\overline{TQ}|\cos(180^\circ - \Theta)$$

$$= |\overline{OP}|^2 + |\overline{OQ}|^2 - 2|\overline{OP}||\overline{OQ}|\cos \gamma$$

$$\tan^2 \alpha + \tan^2 \beta + 2 \tan \alpha \tan \beta \cos \Theta$$

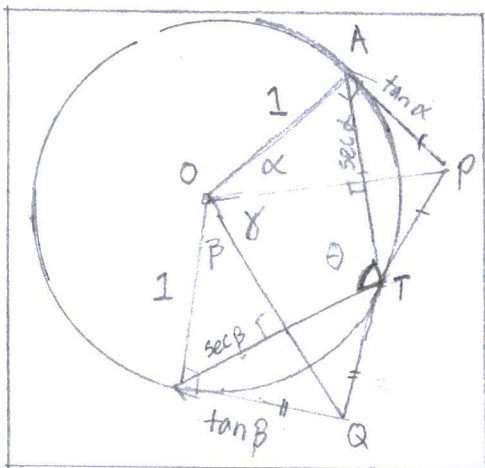
$$= \sec^2 \alpha + \sec^2 \beta - 2 \sec \alpha \sec \beta$$

$$\tan \alpha \tan \beta \cos \Theta = 1 - \sec \alpha \sec \beta \cos \gamma$$

$$\cos \Theta = \frac{1 - \sec \alpha \sec \beta \cos \gamma}{\tan \alpha \tan \beta}$$

$$= \frac{\cos \alpha \cos \beta - \cos \gamma}{\sin \alpha \sin \beta}$$

$$\Theta = \cos^{-1} \left(\frac{\cos \alpha \cos \beta - \cos \gamma}{\sin \alpha \sin \beta} \right)$$



"Internal angle"

Note: Internal angles in Klein and Poincare discs are 90°

Klein model Pentagon:

$$\Theta = \cos^{-1} \left(\frac{\cos\left(\frac{72}{2}\right)\cos\left(\frac{72}{2}\right) - \cos\left(2 \cdot 72 - \left(\frac{72}{2}\right) \times 2\right)}{\sin\left(\frac{72}{2}\right)\sin\left(\frac{72}{2}\right)} \right)$$

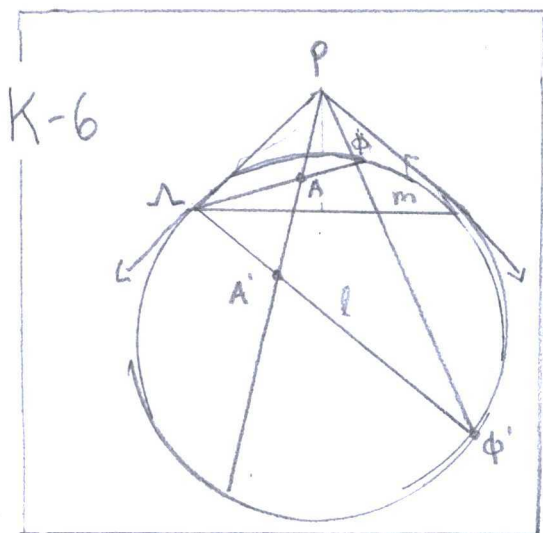
$$= 0.01...$$

$$\cong 0$$

(from exercise) "There exists a hexagon with six right angles."

Klein model Hexagon:

$$\Theta = \cos^{-1} \left(\frac{\cos\left(\frac{60}{2}\right)\cos\left(\frac{60}{2}\right) - \cos\left(2 \cdot 60 - \left(\frac{60}{2}\right) \times 2\right)}{\sin\left(\frac{60}{2}\right)\sin\left(\frac{60}{2}\right)} \right)$$



"Klein Reflection"

(from K-exercise) "Construction of the Klein reflection A' of A across m "

Proof: Hypothesis

Evidence

1) Info given from figure

RAA conjecture

2) $\perp$ is an ideal point to line m

(pg 310) "Definition-Ideal Point"

3) Point P is the pole to m from $\perp$

(pg 314) "Definition-Pole"

4) Points A and A' are interior points

(pg 12) "Definition-points"

5) $\perp$ and A join into a line

Euclid Postulate I

6) Line $\overleftrightarrow{\perp A}$ cuts circle γ at ideal point Φ

(pg 310) "Definition-Ideal Point"

7) Points P and Q cut γ at ideal point Φ'

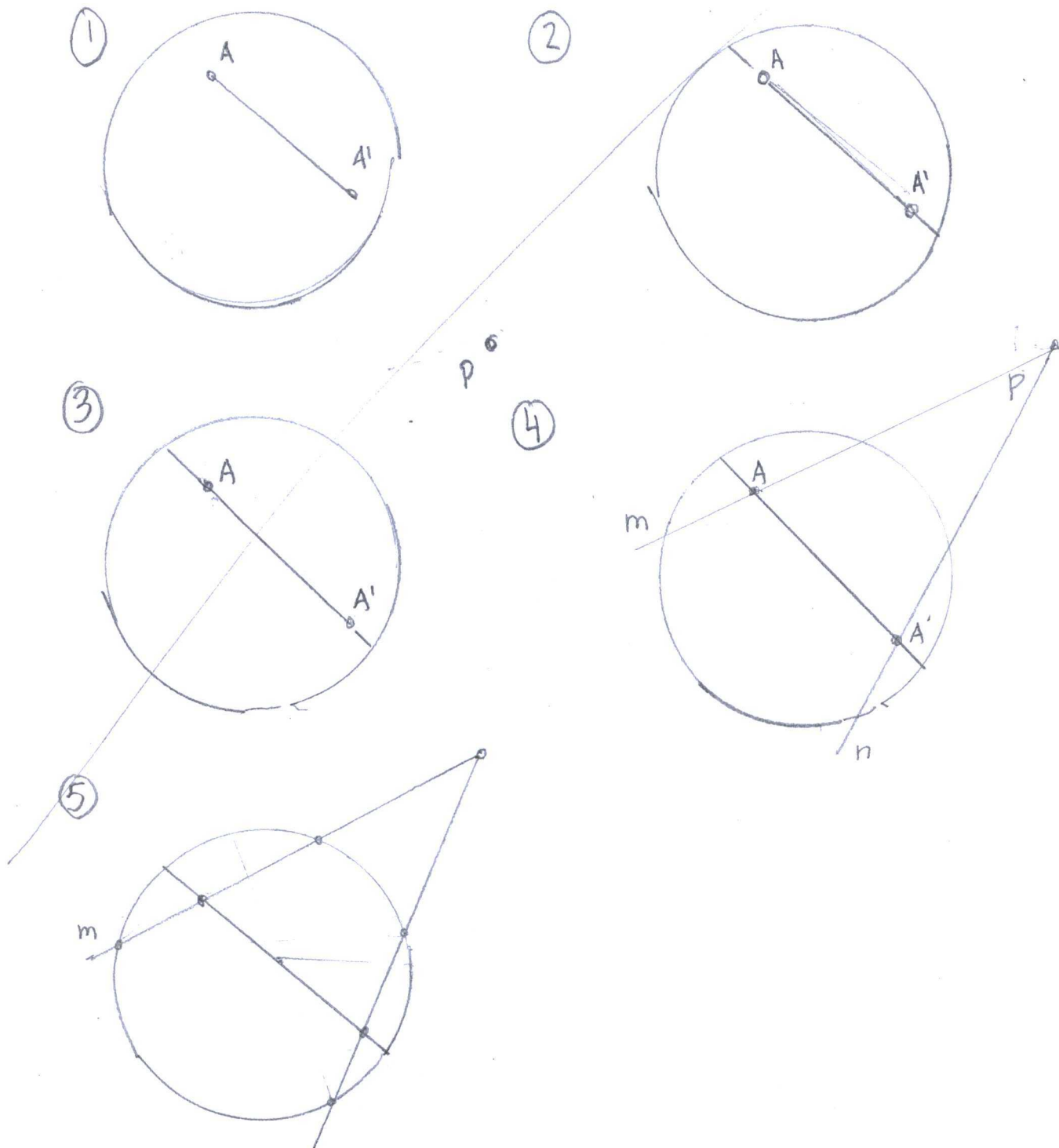
(pg 310) "Definition-Ideal Point"

o) A' is the intersection of $\overline{AP}$ with $\perp \phi'$

Theorem 7.3
and
RAA outcome

K-7.

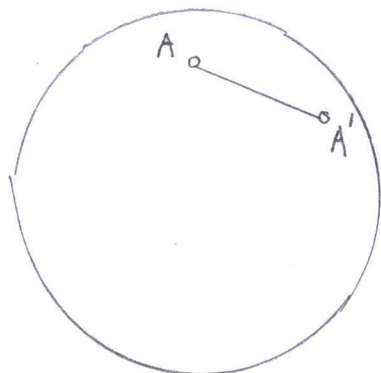
Hyperbolic midpoint(M) by straightedge
and compass:



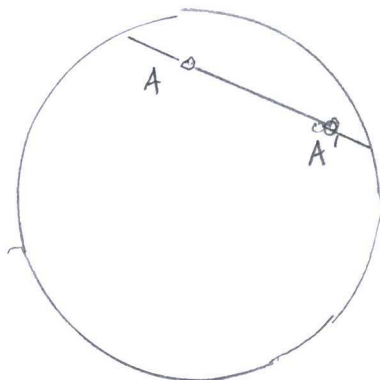
K-7

Hyperbolic midpoint (M) to $\overline{AA'}$ by
straightedge and compass:

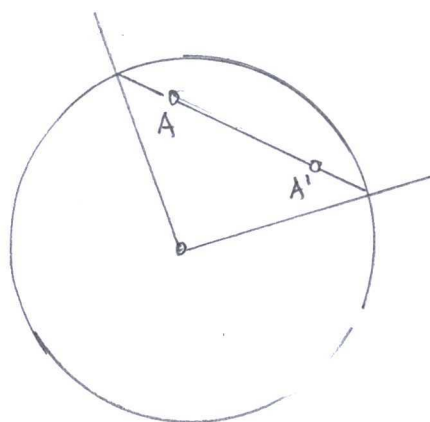
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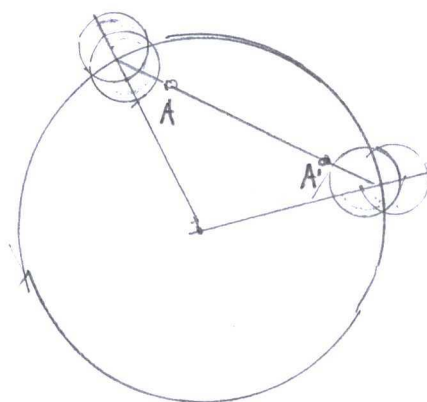
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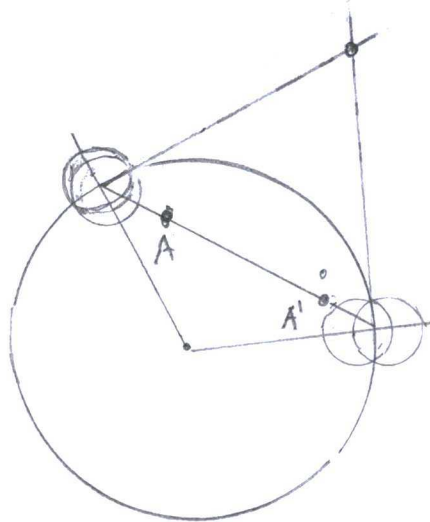


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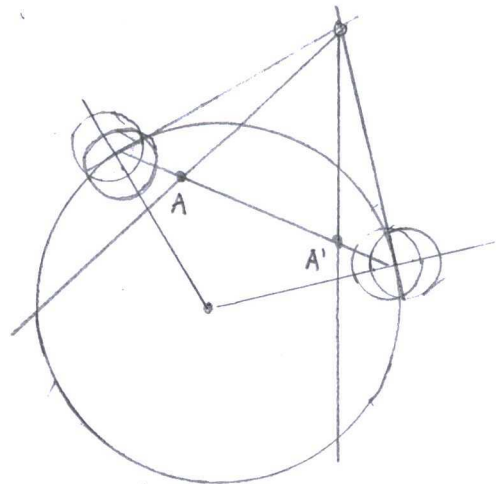


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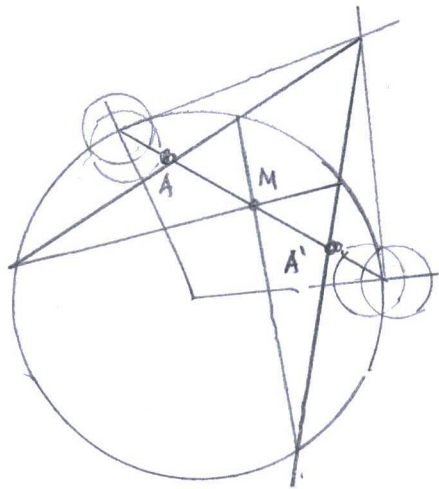
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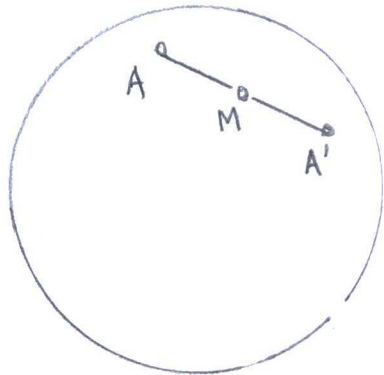
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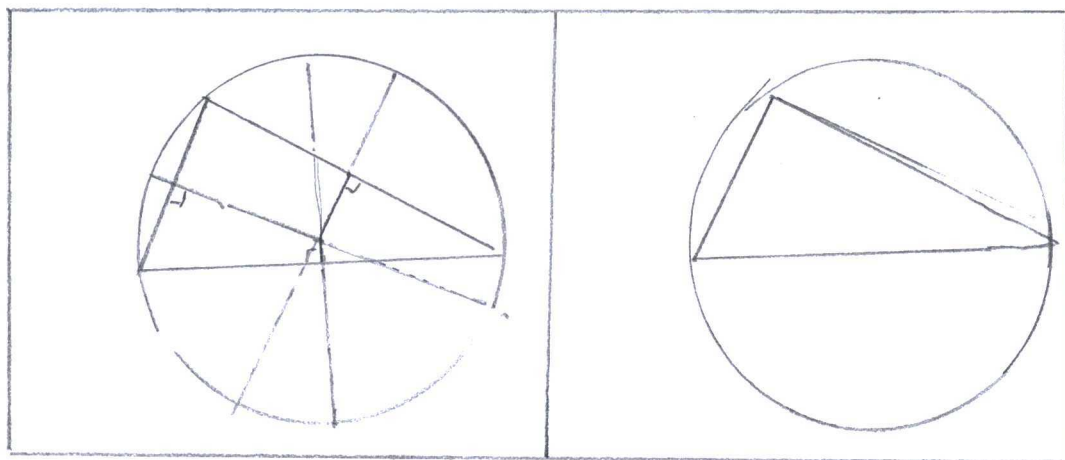


(8)



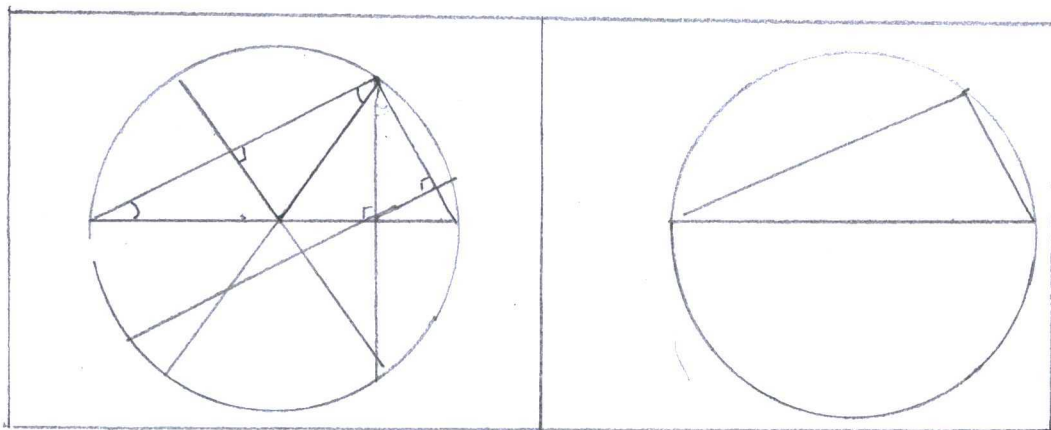
K-8

a) (from K-exercise) "Klein model such that the perpendicular bisectors are divergently parallel."



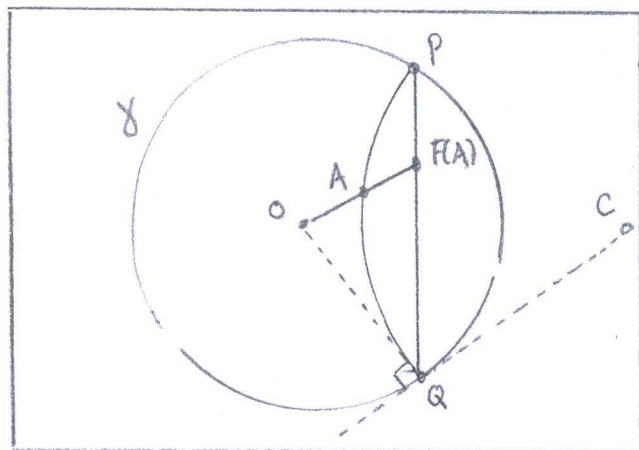
"Perpendicular bisectors are divergently parallel."

b) (from K-exercise) "Klein model such that perpendicular bisectors are ... asymptotically parallel."



"Perpendicular bisectors are asymptotically parallel."

K-9.



(From K-exercise) "Prove the Isomorphism F of the Poincaré model"

$$F(z) = \frac{2z}{1+|z|^2}$$

"Isomorphism F via the sphere"

(pg 332-333) Theorem 7.2

$$e^d = \frac{1 + \tan \beta}{1 - \tan \beta}$$

$$= \frac{1 + OA}{1 - OA}$$

Proof with hyperbolic identities:

$$\sinh(d) = \frac{e^{2d} - 1}{2e^d}$$

$$\cosh(d) = \frac{e^{2d} + 1}{2e^d}$$

$$\tanh(d) = \frac{\sinh(d)}{\cosh(d)}$$

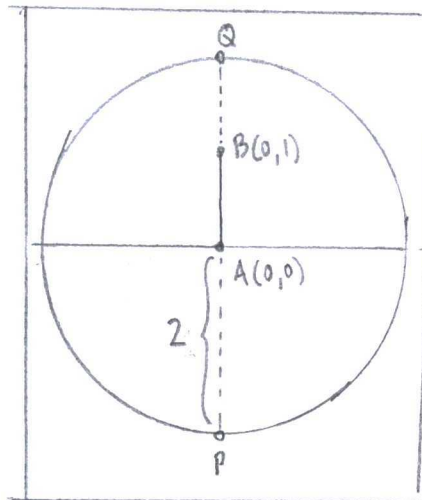
$$= \frac{2 \cdot OA}{1 + OA^2} \approx \frac{2z}{1 + |z|^2} \quad \text{when } OA = z$$

Inverse Isomorphism:

$$F(z) = F^{-1}(z) \text{ because } \frac{z}{1+|z|^2} = \frac{z(1/z)}{1+|1/z|^2}$$

K-10

a) Klein Length $d(AB)$



(pg 319) "Cross ratio"

$$(AB, PQ) = \frac{AP}{BP} \frac{BQ}{AQ}$$

(pg 328) "Distance"

$$e^d = (OB, PQ) = \frac{OP}{OQ} \frac{BQ}{BP}$$

Klein ratio: $e^d = (AB, PQ)$

$$d = |\log_e((AB, PQ))|$$

(from exercise) $A = (0,0)$, $B = (0,1)$

$$d = |\log_e((AB, PQ))|$$

$$= \left| \log_e \left(\frac{2}{2} \cdot \frac{1}{3} \right) \right|$$

$$= \log_e(3)$$

b) Midpoint M on AB:

$$d(A, M, PQ) = \frac{AP}{MP} \frac{MQ}{AQ}$$

$$d(M, B, PQ) = \frac{MP}{BP} \frac{BQ}{MQ}$$

$$= d(A, M, PQ)$$

$$\frac{AP}{MP} \frac{MQ}{AQ} = \frac{MP}{BP} \frac{BQ}{MQ}$$

$$\left(\frac{MQ}{MP}\right)^2 = \frac{AQ}{AP} \frac{BQ}{BP}$$

$$= \frac{2}{2} \cdot \frac{1}{1/3}$$

$$= 3$$

$$MQ = 3 - \sqrt{3}$$

$$M = (0, 3 - \sqrt{3})$$

c) Equidistant curve:

$$a = 1, b = 1/2$$

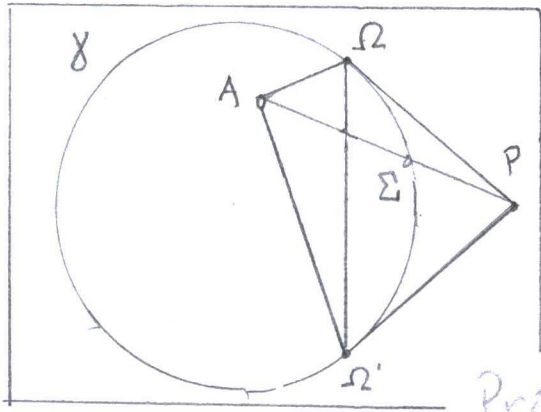
$$y = \sqrt{1 - x^2} / 2$$

Ellipse formula:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

K-11

(from K-exercise) " $A\Sigma$ represents the bisector of $\angle \Omega A \Omega'$ in the Klein model"



"Line of Enclosure"

Proof: Hypothesis

1) Ω and Ω' are distinct ideal points on a circle

2) Point A is an ordinary point

3) $\overrightarrow{AP}$ is a ray through γ at Σ

4) $A\Sigma$ is perpendicular to $\Omega\Omega'$ in Klein model

5) $A\Sigma$ is a perpendicular bisector to $\angle \Omega A \Omega'$

Evidence

(pg 310) "Definition -
Ideal Points"

(pg 310) "Definition - Ordinary Point"

(pg 18) "Definition-
ray"

(pg 309) "Perpendicular
in the Klein model"

(pg 309) "Perpendiculars in Klein model."

in Klein disc.

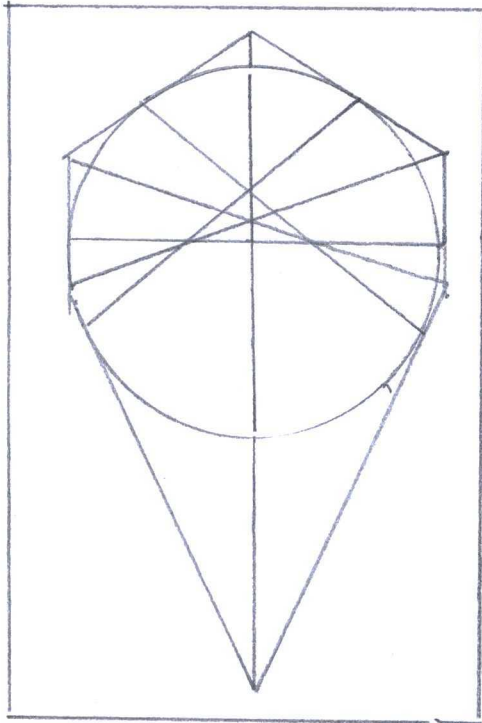
K-12

Cite:

Coxeter, H., Greitzer S.

"Geometry Revisited"

University of Toronto, Rutgers (1967)

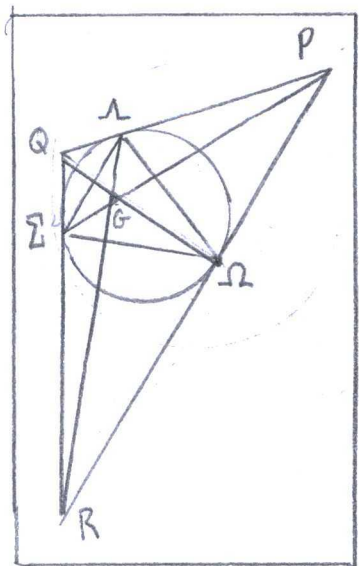


"Brianchon's Theorem"

C. J. Brianchon (1760-1854) constructed an irregular hexagon around a circle. The tangent sides to the circle by the hexagon related to a projection.

K-13

(from K-exercise) "In hyperbolic geometry that inside every trebly asymptotic triangle $\Delta \Sigma \Omega \Lambda$ there is a unique point G equidistant from all sides, which is the point of concurrences of the altitudes."



"Gergonne Point G"

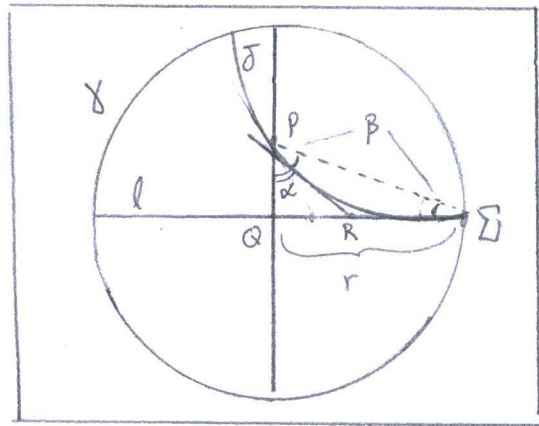
K-14

(From K-exercise) "With suitable ordering of ends P and Q of Klein line through A and B , you have the formula:

$$(AB, PQ) = \frac{a_1 b_1 + a_2 b_2 - 1 - \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_1 b_2 - a_2 b_1)^2}}{a_1 b_1 + a_2 b_2 - 1 + \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_1 b_2 - a_2 b_1)^2}}$$

K-15

(from K-exercise) "Proof of the Bolyai -
Lobachevsky formula"



(pg 343) Klein Length

$$d'(AB) = |\log(AB, PQ)|$$

(pg 332) "Bolyai-Lobachevsky Formula"

$$= \frac{1+AB}{1-AB}$$

$$AB = \frac{e^d - 1}{e^d + 1} = \tanh\left(\frac{d}{2}\right)$$

$$\begin{aligned} \text{Thus, } e^{-d} &= \frac{1 - \tanh(d/2)}{1 + \tanh(d/2)} \\ &= 1 - \frac{1 - \tanh(d/2)}{1 + \tanh(d/2)} \\ &= \frac{1 + \tanh(d/2) - 1 + \tanh(d/2)}{1 + \tanh(d/2) + 1 - \tanh(d/2)} \\ &= \tanh(d/2) \end{aligned}$$

K-16.

a) (From K-exercise) "Cartesian Line"

$$Ax + By + C = 0$$

Cartesian lines are secant to a unit circle ($x^2 + y^2 = 1$) when $A^2 + B^2 - C^2 > 0$.

Why? The line has two solutions.

When $A^2 + B^2 - C^2 > 0$ crosses the perimeter twice.

b) (from K-exercise) "If $P' = (x', y')$ is the Klein reflection of point $P = (x, y)$ across l , then

$$x' = \frac{|l|^2 x - 2A(Ax + By + C)}{|l|^2 + 2C(Ax + By + C)}$$

$$y' = \frac{|l|^2 y - 2B(Ax + By + C)}{|l|^2 + 2C(Ax + By + C)}$$

Proof: Hypothesis

Case 1: $C = 0$

1) $P(x, y) \xrightarrow[\text{across } Ax + By = 0]{\text{reflects}} P'(x', y')$

Evidence

Theorem 7.3

Euclidean Reflection

$$x' = x - \frac{A(Ax + By)}{A^2 + B^2} = \frac{|l|^2 x - 2A(Ax + By)}{A^2 + B^2}$$

$$y' = y - \frac{B(Ax + By)}{A^2 + B^2} = \frac{|l|^2 y - 2B(Ax + By)}{A^2 + B^2}$$

Case 2: $C \neq 0$

1) Pole L is at $(-A/C, -B/C)$

(pg 314) "Definition-Pole"

2) Midpoint M is the

(pg 326) Theorem on

intersection between LP and l

3) A random line n through $P(x, y)$ meets Σ at Σ' and Σ''

4) Two temporary lines ΣM and $\Sigma' M$ meet at Σ at ideal points Ω and Ω' .

5) $P'(x', y')$ is the intersection of $N\Omega'$ and $L\Omega$

$$x' = \frac{|l|^2 x - 2A(Ax + By + C)}{|l|^2 + 2C(Ax + By + C)}$$

$$y' = \frac{|l|^2 y - 2B(Ax + By + C)}{|l|^2 + 2C(Ax + By + C)}$$

the construction of P-reflections

Proposition 2.1

(pg 310) "Definition - Ideal points"

Proposition 2.1

c) Klein Perpendiculars:

Line	Standard Form	Slope-intercept
l	$Ax + By + C = 0$	$y = \frac{-A}{B}x - \frac{C}{B}$
l'	$A'x + B'y + C' = 0$	$y = \frac{-A'}{B'}x - \frac{C'}{B'}$

Perpendicular Line:

$$\text{slope (@ } c=0): \quad \frac{-A}{B} \cdot \frac{-A'}{B'} = -1$$

$$AA' + BB' = 0$$

$$\text{slope (@ } c \neq 0): \quad \frac{-A}{B} \left(\frac{-A'}{B'} \right) = -1 + \frac{CC'}{BB'}$$

$$AA' + BB' - CC' = 0$$

d) (From K-exercise) "Determinant Criteria for... parallel"

$$D = \det \begin{pmatrix} A & B \\ A' & B' \end{pmatrix} = AB' - BA' = 0$$

Klein Parallel sufficient condition:

Point at intersection:

$$\begin{array}{c|c} \text{X-coordinate} & \text{Y-coordinate} \\ \hline \left(\frac{A}{B}\right)x - \frac{C}{B} = \left(\frac{-A'}{B'}\right)x - \frac{C'}{B'} & \left(\frac{-B}{A}\right)y - \frac{C}{A} = \left(\frac{-B'}{A'}\right)y - \frac{C'}{A'} \end{array}$$

$$\left[\frac{-A}{B} + \frac{A'}{B'} \right] X = \frac{C}{B} - \frac{C'}{B'}$$

$$y = \frac{AC' - A'C}{D}$$

$$\frac{-AB' + A'B}{BB'} X = \frac{CB' - BC'}{BB'}$$

$$\underbrace{-D} \cdot X = CB' - BC'$$

$$X = \frac{BC' - B'C}{D}$$

In a circle ($r=1$), $X^2 + y^2 = 1$

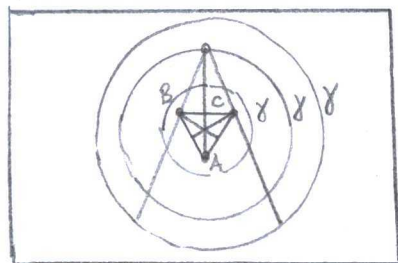
$$\left(\frac{BC' - B'C}{D} \right)^2 + \left(\frac{AC' - A'C}{D} \right)^2 = 1$$

$$(BC' - B'C)^2 + (AC' - A'C)^2 \leq D^2$$

...which is a sufficient condition for intersection in a unit circle.

K-17

a) (from K-exercise) "The internal bisector of $\angle A$ is concurrent with the external bisectors of $\angle B$ and $\angle C$."



b) Ordinary Vertices in a concurrent triangle $\triangle ABC$:

The external bisector to $\angle A$ has equivalent length as $\angle B$ and $\angle C$. Also, bisectors are not parallel.

Ideal Vertices in a concurrent triangle $\triangle ABC$:

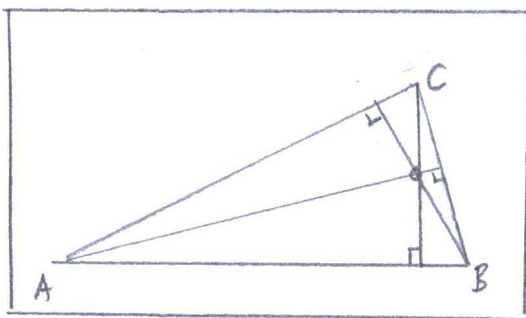
Center I of circle γ is center to $\triangle ABC$. Vertices A, B , and C lay on circle γ .

Ultra-Ideal vertices in a concurrent triangle $\triangle ABC$:

The ultra-ideal points intersect γ from point-pairs $A-B, B-C$, and $A-C$.

K-13

Ordinary Vertices of an acute triangle $\triangle ABC$:



"Altitudes to an acute triangle are concurrent"

The shape is an acute triangle because every exterior angle more than remote interior.

An Exterior Angle (EA) Theorem 4.2 supports an acute triangle in Klein Discs. Altitudes to $\triangle ABC$ intersect opposite sides by crossbar theorem (pg 116).

Ideal vertices of an acute triangle

$\triangle ABC$:

Vertices A, B, C coincide with circle δ . The triangle is acute because Exterior Angle Theorem 4.2. Crossbar theorem deduces concurrent altitudes.

Ultra-ideal vertices in an acute triangle $\triangle ABC$:

Sides AB, BC , and AC each have

Desargues argument (pg 98) describes a relationship between two triangles at the vertices and along the sides. Similar triangles have concurrent medians.

b) (personal words) Converse to Desargues Theorem

"Collinear sides correspond, if the vertices between two triangles are concurrent."

Proof Hypothesis	Evidence
1) A circle γ has a center point O	Euclid's Postulate III
2) Triangle $\triangle ABC$ is in circle γ	(pg 98) "Definition-Triangle"
3) Midpoint to AB is point O , for $A \times O \times B$.	Proposition 4.3 (Midpoints)
4) Two other midpoints reside on $\triangle ABC$: P and Q for $A \times P \times C$ and $C \times Q \times B$	Proposition 4.3 (midpoints)

5) $(AB, PQ) = 1$ is
a harmonic
conjugate to
 PQ .

6) PQ meets AB
at O .

7) Because $\overline{PO}$,
triangle $\triangle ABC$
is concurrent to
 $\triangle OPQ$

8) Vertices in $\triangle ABC$
correspond to $\triangle IJK$

(pg 339) "Definition-
Harmonic
Conjugate"

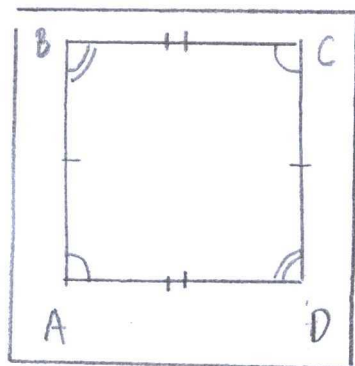
Lemma 7.5

RAA hypothesis

RAA outcome

c) (from K-exercise) "Perpendiculars are
also concurrent in the projective
completion."

K-20



"Symmetric
parallelogram"

(from K-exercise) "Both pairs of opposite angles are congruent."

Proof: Hypothesis	Evidence
1) A quadrilateral $\square ABCD$	(pg 44) "Definition-Quadrilateral"
2) Opposite sides are congruent	Axiom C-2
3) Opposite angles are congruent	Axiom C-2.

(from K-exercise) "Lines containing opposite sides have a common perpendicular."

Proof: Hypothesis	Evidence
1) A quadrilateral $\square ABCD$	(pg 44) "Definition-Quadrilateral"
2) Opposite sides are congruent	Axiom C-2
3) Transverse lines are perpendicular	Proposition 4.1

(from K-exercise) $\square ABCD$ is a symmetric parallelogram iff the diagonals bisect each other"

Proof: Hypothesis	Evidence
1) A quadrilateral $\square ABCD$	(pg 44) "Definition-Quadrilateral"
2) Opposite sides are congruent	Axiom C-2
3) A segment $\overline{AC}$ and $\overline{BD}$ on sides	(pg 14) "Definition-segment"
4) $\angle CAB$ and $\angle ABD$ have bisectors	Proposition 4.4 (Bisectors)
5) Midpoints on $\overline{BC}$ and $\overline{AD}$ are the same point	Proposition 4.3 (Midpoints)

(from K-exercise) "If S is a common midpoint, $\square ABCD$ has an inscribed circle with center S ."

Proof Hypothesis	Evidence
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1) Quad-ril-at-eral

$\square ABCD$

2) Opposite sides are
congruent

3) Sides are $\overline{AC}$ and
 $\overline{BD}$

4) $\overline{AB} = \overline{AC}$

5) S is a common
midpoint on $\overline{AD}$
and $\overline{BC}$

6) $\angle ASB + \angle ASC = 180^\circ$
 $\angle ASC + \angle CSD = 180^\circ$

7) $\angle CSD + \angle DSB = 180^\circ$
 $\angle DSB + \angle ASB = 180^\circ$

7) $90^\circ = \angle ASB = \angle ASB$
 $= \angle CSD = \angle DSB$

8) Midpoint E on
 AB

9) $\overline{SE}$ inscribes a
circle around S

(pg 44) "Definition-
Quadrilateral"

Axiom C-2

(pg 18) "Definition-
sides"

RAA hypothesis

Proposition 4.3 (Midpoints)

(pg 19) "Definition-
Supplementary"

Proposition 3.19

Proposition 4.3 (Midpoints)

Euclid's Postulate III
and RAA result

(from K-exercise) "Diagonals of a symmetric parallelogram are congruent iff all four angles are congruent."

Proof: Hypothesis	Evidence
1) Quadrilateral $\square ABCD$	(pg 44) "Definition-Quadrilateral"
2) Opposite sides are congruent	(pg 356) "Definition-Parallelogram?"
3) Sides are $\overline{AC}$ and $\overline{BD}$	(pg 18) "Definition-Sides"
4) $\overline{AB} = \overline{AC}$	RAA hypothesis
5) S is a common midpoint	Proposition 4.3 (Midpoints)
6) $\triangle ASB \cong \triangle CSD$	Proposition 3.22 (S.S.S.)
7) $\angle ASB \cong \angle ASC$ $\cong \angle CSD \cong \angle DSB$	(pg 14) "Definition-congruent"
8) $\overline{AD} \cong \overline{CB}$	RAA result

b) (from K-exercise) "Every parallelogram in a hyperbolic plane, with S

as the midpoint of its diagonals"

Proof: Hypothesis	Evidence
<u>Case 1: $\overline{AB} = \overline{CD}$</u>	
1-5) Previous proof	
<u>Case 2: $\overline{AB} \neq \overline{CD}$</u>	
1) $\square ABCD$	(pg 44) "Definition - Quadrilateral"
2) Opposite sides are congruent	(pg 14) "Definition - congruent"
3) Sides are $\overline{AC}$ and $\overline{BD}$	(pg 18) "Definition - sides"
4) S is a common midpoint on $\overline{AD}$ and $\overline{CB}$	Proposition 4.3 (Midpoints)

c) (From K-exercise) " $\square ABCD$ is a Euclidean parallelogram, that is a Euclidean rectangle iff all four angles are Klein-congruent, and that it is a Euclidean square iff it is a hyperbolic regular 4-gon."

Proof: Hypothesis

Evidence

1) $\square ABCD$ is a symmetric Euclidean parallelogram with S the midpoint of its diagonals.

(pg 356) "Definition-Symmetric Parallelogram"

2) $\angle CAB \cong \angle ABD$
 $\cong \angle BDC \cong \angle DCA$

Internal angles in a Klein Disc. from exercise K-5

$$\cong \cos^{-1} \left(\frac{\cos(\frac{90}{2}) \cos(\frac{90}{2}) - \cos(2 \cdot 90 - \frac{90}{2} \cdot 2)}{\sin(\frac{90}{2}) \sin(\frac{90}{2})} \right)$$

$\cong 0^\circ$ in a Klein disc.

(from K-exercise) "Suppose $\square ABCD$ (from above), then each pair of opposite sides, S is a symmetry point for the lines containing those sides."

Proof: Hypothesis

Evidence

1) Information above...

RAA hypothesis

2) Midpoints E, F, G, H on $\overline{AB}, \overline{AC}, \overline{CD},$ and $\overline{DB}.$

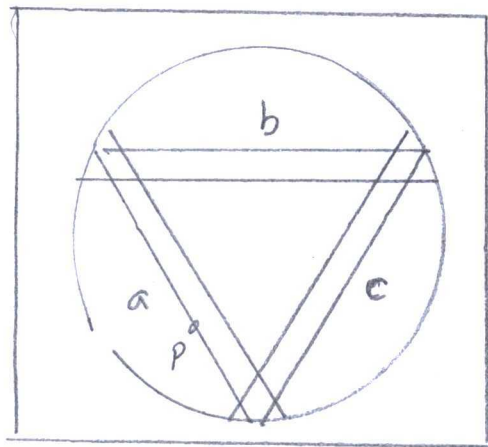
(pg) "Definition-midpoints"

$$12) \bar{E}S = GS = FS = HS$$

RAA result

d) A proof about an inner triangle $\triangle ABC$ needs midpoint theories, concurrent definitions: bisectors, and measurement theorems.

K-21,



cite: Blumenthal, L., Menger, K.
"Studies in Geometry"
University of Missouri (1970)

The guys found an argument
Unbeknownst between Klein and
Euclidean fields.